

L1.8 Linearization & Taylor

§2.9, §11.10 — what you need to know

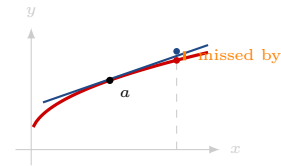
LINEARIZATION

$$L(x) = f(a) + f'(a)(x - a)$$

“the tangent line, used as a calculator”

Two ingredients, both evaluated **at** a : the height and the slope. Choose a close to the value you want, and somewhere you can evaluate f exactly.

Small angle. At $a = 0$: $\sin x \approx x$, $\cos x \approx 1$, $\tan x \approx x$ for $x \lesssim 10^\circ$. **Radians only** — it sets an angle equal to a length.



ACTUAL CHANGE AND THE DIFFERENTIAL

$$\Delta y = f(x + \Delta x) - f(x) \quad dy = f'(x) dx$$

“ Δy is what happened; dy is what the line predicted”

Take $dx = \Delta x$ so both describe the same step in. The gap between them is exactly **how far the tangent line’s estimate missed**, and it closes far faster than the step does — halve dx and the gap drops by about four.

FRACTIONAL UNCERTAINTY

A measurement arrives as a value and an **uncertainty**, $r = 21 \pm 0.05$ cm. That uncertainty **is** a dr , and anything computed from it inherits one: $dV = f'(r) dr$. *Not the same thing as the gap above* — that was how far an estimate missed; this is how well a measurement is known.

$$\frac{df}{f} = n \left(\frac{dx}{x} \right) \quad \text{for } f = Cx^n$$

“the exponent becomes the multiplier”

A 1% uncertainty in a radius is 3% in a volume, 2% in an area, 0.5% in a pendulum’s period.

Report both — the absolute uncertainty carries units, the fractional one transfers: $V = 38,792 \pm 277$ cm³, $\frac{dV}{V} = 0.71\%$.

SUMMATION CLOSED FORMS

$\sum_{i=1}^n a_i = a_1 + a_2 + \dots + a_n$, where i is the **index**. Given, not derived — and the third is the square of the first.

$$\sum_{i=1}^n i = \frac{n(n+1)}{2} \quad \sum_{i=1}^n i^2 = \frac{n(n+1)(2n+1)}{6} \quad \sum_{i=1}^n i^3 = \left[\frac{n(n+1)}{2} \right]^2$$

TAYLOR AND MACLAURIN SERIES

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!} (x-a)^n \quad a=0 \Rightarrow \text{Maclaurin}$$

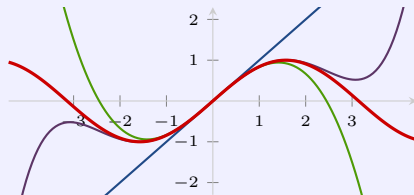
“differentiate n times, evaluate at a , divide by $n!$ ”

a is **where you aim it** — same function, different centre, different series, each best near its own. The three worth knowing cold, all at $a=0$:

$$e^x = 1 + x + \frac{x^2}{2!} + \dots \quad \sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots \quad \cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \dots$$

The truncation ladder

One formula, cut off at different places. Drawn on $\sin x$ about $a=0$, whose even terms are all zero.



$\sin x$ the function
 $T_1 = x$ this is $L(x)$

$$T_3 = x - \frac{x^3}{3!}$$

$$T_5 = x - \frac{x^3}{3!} + \frac{x^5}{5!}$$

All of them are exact at a and get worse as you walk away. More terms buys *more room*, not accuracy everywhere.

Keep the $n=0$ and $n=1$ terms of the Taylor series and you get $f(a) + f'(a)(x-a)$. **Linearization is the degree-1 Taylor polynomial** — the two halves of this lesson are one idea at two truncations.

WATCH OUT

- ✗ ~~linearization is a new idea~~ → it is the tangent line you already had, given a new job
- ✗ ~~Δy and dy are the same~~ → Δy happened, dy was predicted — the gap is how far the estimate missed
- ✗ ~~the uncertainty in V is the uncertainty in r~~ → it is three times the fractional uncertainty, since $V \propto r^3$
- ✗ ~~$\sin x \approx x$ always~~ → only for small x , and only in radians
- ✗ ~~a Maclaurin series works everywhere~~ → it is built at $a=0$ and is best *near* $a=0$
- ✗ ~~more terms is always better~~ → it helps near a ; far from a it can still be useless